

Question	Answer	Marks
3(a)	<u>sum/total</u> momentum (of a system of bodies) is constant or <u>sum/total</u> momentum before = <u>sum/total</u> momentum after	M1
	for an isolated system or no (resultant) external force	A1
3(b)(i)	$m = \rho V$	C1
	$= 1.3 \times \pi \times 0.045^2 \times 1.8 \times 2.0 = 0.030 \text{ (kg)}$	A1
3(b)(ii)	1. $(\Delta)p = (\Delta)mv$	C1
	$= 0.030 \times 1.8$ $= 0.054 \text{ N s}$	A1
	2. $F = 0.054/2.0$ or $0.030 \times 1.8/2.0$ $= 0.027 \text{ N}$	A1
3(b)(iii)	force on air (by propeller) equal to force on propeller (by air)	M1
	and opposite (in direction)	A1
3(b)(iv)	resultant force = 0.20×0.075 (= 0.015 N) frictional force = $0.027 - 0.015$	C1
	$= 0.012 \text{ N}$	A1

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1(c)	vibration(s)/oscillation(s) (of particles) parallel to direction of propagation of energy	D1